

Wireless Security (Part-3)

- Topics covered:
 - Intro to Wireless Network Security
 - Authentication Methods
 - Encryption/Integrity Methods
 - Wi-Fi Protected Access (WPA)

Intro to Wireless Network Security

Wireless Network Security

This section introduces the three core concepts of wireless network security:

1. Authentication
2. Encryption
3. Integrity

It also introduces **Wi-Fi Protected Access (WPA)**, which provides standardized wireless security protocols.

Why Wireless Security Is Important

Security is important in all networks, but it is especially important in wireless networks because wireless signals are not contained within a cable.

Any device within range of the wireless signal can receive the transmitted traffic.

In wired networks, traffic is often encrypted only when it crosses an untrusted network such as the Internet. Traffic inside the LAN is usually not encrypted.

In wireless networks, traffic exchanged between wireless clients and the Access Point (AP) should be encrypted because any nearby device can receive the signal. Encryption ensures that only the intended recipient can read the traffic.

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Authentication

Authentication is the process of verifying the identity of a user and/or device.

Before a client can associate with an AP, it must first be authenticated.

Authentication in Different Environments

- In corporate networks, only trusted users and devices should be granted access.
- Guest users can be provided with a separate SSID.
- Guest SSIDs usually has less strict authentication requirements.
- Guest networks typically provide Internet access only and not access to internal company resources.

Mutual Authentication

Ideally:

1. The AP should authenticate clients.
2. Clients should also authenticate the AP.

This helps prevent association with a malicious AP.

A malicious AP may trick users into connecting and then perform attacks such as a **man-in-the-middle attack**.

Authentication Methods

Authentication can be performed using:

1. Passwords
2. Username and password combinations
3. Digital certificates installed on devices

Authentication Process

Before association, a client must complete an authentication process.



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Encryption

Wireless traffic exchanged between clients and APs should be encrypted.

Encryption prevents anyone except the sender and intended recipient from reading the traffic.

Encryption works by scrambling a message so that only authorized parties can read it.

Encryption Requirements

Many encryption protocols exist.

The sender and receiver must use the same encryption protocol. Otherwise, the recipient will be unable to decrypt the sender's messages.

Encryption Keys in Wireless LANs

All devices on a wireless LAN use the same encryption protocol.

However:

- Each client uses a unique encryption/decryption key.
- Other clients cannot read that client's traffic.
- The AP possesses the appropriate key needed to decrypt each client's traffic.

Group Key

- The AP also maintains a **group key**.
- This key is used when the AP sends traffic intended for all clients.
- All clients store a copy of the group key so they can decrypt those messages.

Integrity

Integrity ensures that messages are not modified by a third party while in transit.

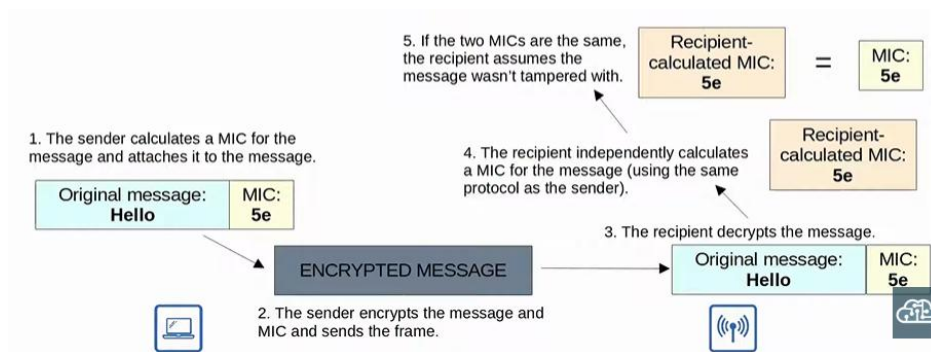
The message received by the destination should be identical to the message sent by the source.

To help verify integrity, a **MIC (Message Integrity Check)** is added to messages.

MIC Operation Process

1. The sender calculates a MIC and attaches it to the message.
2. The sender encrypts both the message and MIC and sends the frame.
3. The recipient decrypts the message.
4. The recipient independently calculates its own MIC using the same protocol.
5. The two MIC values are compared.

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Important Notes

- Sender and receiver must use the same MIC calculation protocol.
- If the two MIC values match, the recipient assumes the message was not tampered with.
- If the MIC values do not match, the message is discarded.

Therefore, a MIC helps determine whether message integrity has been compromised.



Authentication Methods

Wireless Authentication Methods

Several authentication methods exist.

Only a basic understanding of each method is required, but they can be easy to confuse when first learning them.

The original IEEE 802.11 standard included two authentication options:

1. Open Authentication
2. WEP (Wired Equivalent Privacy)

Open Authentication

Open authentication is extremely simple.

The client sends an authentication request and the AP accepts it without requiring credentials.

Characteristics

1. No credentials are required.
2. Not a secure authentication method.
3. The AP accepts all authentication requests.

Modern Usage

Open authentication is still used together with other authentication methods.

For example:

- A user may connect to a public Wi-Fi network without entering a password.
- After association, the user may be required to authenticate through a web page.
- Network access is granted only after completing that additional authentication.

Open authentication itself is not secure, but it can be combined with other methods.

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WEP (Wired Equivalent Privacy)

WEP provides both:

1. Authentication
2. Encryption

Encryption Algorithm

WEP uses the **RC4** algorithm for encryption.

Shared-Key Protocol

WEP is a shared-key protocol.

Both sender and receiver must possess the same key.

Key Lengths

WEP keys can be: 40 bits or 104 bits

These are combined with a 24-bit Initialization Vector (IV), resulting in:

1. 64-bit total length
2. 128-bit total length

Security Status

Although longer keys are generally more secure, WEP is not secure.

WEP can be easily cracked regardless of key length and should not be used in modern wireless networks.

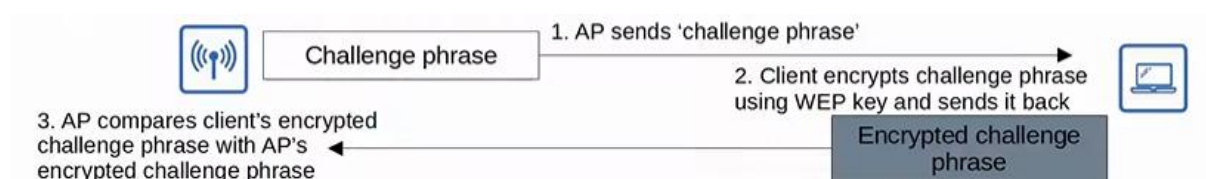
WEP Authentication Process

The AP verifies whether the client knows the shared WEP key.

Process

AP sends a challenge phrase.

1. Client encrypts the challenge phrase using the WEP key and returns it.
2. AP compares the client's encrypted phrase with its own encrypted version.
3. If they match, authentication succeeds.



Key Idea: The AP is simply testing whether the client knows the shared key.

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WEP Usage Options

WEP can be used:

1. For encryption only
2. For both encryption and authentication

If WEP authentication is not used:

- Open authentication is used.
- WEP only provides encryption.

EAP (Extensible Authentication Protocol)

EAP is not a single authentication protocol. It is an **authentication framework** that provides a standard set of authentication functions used by various **EAP methods**.

Examples of EAP methods include:

1. LEAP
2. EAP-FAST
3. PEAP
4. EAP-TLS

EAP is integrated with **802.1X**, which provides **port-based network access control**.

802.1X Overview

802.1x is used to limit network access for clients connected to a LAN or WLAN until they authenticate. There are three main entities in 802.1X:

Supplicant: The device that wants to connect to the network.

Authenticator: The device that provides access to the network.

Authentication Server (AS): The device that receives client credentials and permits/denies access.



Authentication Flow in 802.1X:

1. The client performs **open authentication** and associates with the AP.
2. The client is associated but does not yet have full network access.
3. Only traffic required for EAP authentication is allowed.
4. The Authentication Server checks the client's credentials.
5. The Authentication Server decides whether access is permitted or denied.
6. The WLC acts as a middle-man in the authentication process.

Key Point: Association with the AP uses open authentication. And Full network access requires an additional EAP authentication step.

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1. LEAP (Lightweight EAP)

LEAP was developed by Cisco as an improvement over WEP.

Authentication Process

Clients must provide: Username and Password

LEAP also provides *mutual authentication*.

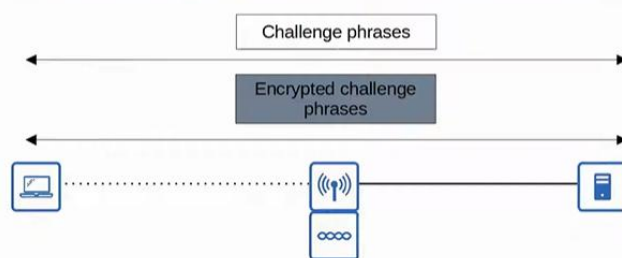
Unlike WEP, where only the server sends a challenge phrase, both the client and server send challenge phrases.

Process

1. Challenge phrases are exchanged.
2. Each device encrypts the other device's challenge phrase.
3. The encrypted challenge phrases are returned.
4. Both devices authenticate each other.

LEAP (Lightweight EAP)

- LEAP was developed by Cisco as an improvement over WEP.
- Clients must provide a username and password to authenticate.
- In addition, *mutual authentication* is provided by both the client and server sending a challenge phrase to each other.
- *Dynamic* WEP keys are used, meaning that the WEP keys are changed frequently.
- Like WEP, LEAP is considered vulnerable and should not be used anymore.



Additional Security Feature

LEAP uses *dynamic WEP keys*.

These keys automatically change over time, making encryption more difficult to crack.

Security Status: LEAP is considered vulnerable and should not be used.

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2. EAP-FAST

EAP Flexible Authentication via Secure Tunneling was also developed by Cisco.

It consists of three phases.

Phase 1 – PAC Provisioning

A PAC (*Protected Access Credential*) is generated and passed from the server to the client.

The PAC acts like a shared key.

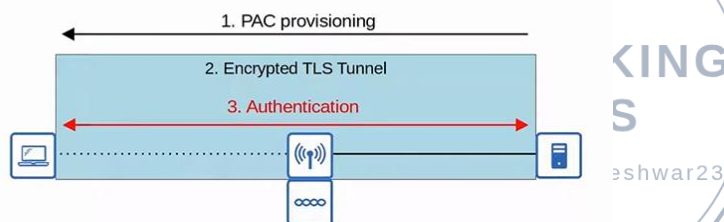
Phase 2 – TLS Tunnel Creation

The PAC is used to establish a secure TLS tunnel between the client and the Authentication Server.

This tunnel allows secure encrypted communication.

Phase 3 – Authentication

The client is authenticated inside the TLS tunnel.



3. PEAP (Protected EAP)

PEAP is similar to EAP-FAST.

Like EAP-FAST, it establishes a secure TLS tunnel between the client and server.

The difference is how the tunnel is established.

(PEAP Process)

Step 1 – Digital Certificate

The server presents a digital certificate.

The client uses the certificate to authenticate the server.

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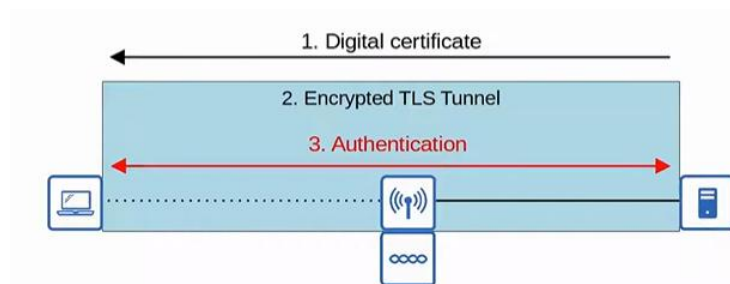
Step 2 – TLS Tunnel

The certificate is used to establish a secure TLS tunnel.

Step 3 – Client Authentication

The client is authenticated inside the TLS tunnel.

One protocol that can be used for this authentication is: MS-CHAP (Microsoft Challenge Handshake Authentication Protocol)



Key Point:

1. EAP-FAST uses a PAC
2. PEAP uses a digital certificate

Both methods:

- Establish a secure TLS tunnel
- Authenticate the client inside that tunnel

4. EAP-TLS

EAP Transport Layer Security is considered the most secure authentication method.

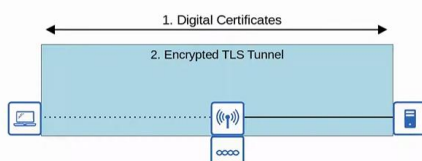
Certificate Requirements

Unlike PEAP:

- The Authentication Server must have a certificate.
- Every client must also have a certificate.

EAP-TLS (EAP Transport Layer Security)

→ Whereas PEAP only requires the AS to have a certificate, EAP-TLS requires a certificate on the AS and on every single client.
→ EAP-TLS is the most secure wireless authentication method, but it is more difficult to implement than PEAP because every client device needs a certificate.
→ Because the client and server authenticate each other with digital certificates, there is no need to authenticate the client within the TLS tunnel.
→ The TLS tunnel is still used to exchange encryption key information (encryption methods will be discussed next!).



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Authentication Process

Because both client and server authenticate each other using digital certificates:

No separate client authentication inside the TLS tunnel is required.

The TLS tunnel is still used to exchange encryption key information.

Advantages: Most secure authentication method

Disadvantages

1. More difficult to deploy
2. Every client requires a certificate
3. Adds complexity, time, and effort

Summary: Many enterprises prefer PEAP because it is easier to implement, even though EAP-TLS provides stronger security.

Authentication Methods

Open Authentication

WEP (Wired Equivalent Privacy)

EAP (Extensible Authentication Protocol)

LEAP (Lightweight EAP)

EAP-FAST (EAP Flexible Authentication via Secure Tunneling)

PEAP (Protected EAP)

EAP-TLS (EAP Transport Layer Security)

Encryption/Integrity Methods

Wireless Encryption Methods

After authentication, encryption protects wireless traffic and keeps data private.

The encryption methods discussed are:

1. TKIP
2. CCMP
3. GCMP

(WEP was already discussed previously.)

1. TKIP (Temporal Key Integrity Protocol)

TKIP was developed after WEP was found to be vulnerable.

At the time, existing wireless hardware was built for WEP, so a temporary solution was needed until a new standard and new hardware became available.

TKIP served as that temporary solution.

It is based on WEP but adds several security improvements.

(TKIP Security Enhancements)

1. MIC Protection

TKIP uses a *MIC* (*Message Integrity Check*) to protect message integrity.

2. Key Mixing Algorithm

A key mixing algorithm generates a unique WEP key for every frame.

Instead of all frames using the same key, each frame uses a unique key.

This makes encryption harder to crack.

3. Larger Initialization Vector

The initialization vector is increased:

WEP: 24 bits

TKIP: 48 bits

This makes brute-force attacks much more difficult.

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4. Sender MAC Address Included

The MIC includes the sender's MAC address.

This helps prove the identity of the frame sender.

5. Timestamp Protection

A timestamp is added.

This helps prevent replay attacks.

6. TKIP Sequence Number

TKIP keeps a sequence number for frames sent from a source MAC address.

Frames with incorrect sequence numbers are dropped.

This also protects against replay attacks.

TKIP Whole

TKIP was developed as a more secure temporary replacement for WEP.

Key characteristics:

1. Based on WEP
2. Uses MIC for integrity
3. Uses unique keys for frames
4. Uses a 48-bit initialization vector
5. Includes sender MAC address verification
6. Uses timestamps
7. Uses sequence numbers to prevent replay attacks

TKIP is the encryption method used in **WPA (Wi-Fi Protected Access Version 1)**.

2. CCMP (Counter/CBC-MAC Protocol)

CCMP was developed after TKIP and provides stronger security.

It is used in **WPA2**.

A device must have hardware support for CCMP. Older hardware designed only for WEP or TKIP cannot use CCMP.

CCMP combines two algorithms:

1. Encryption
2. Message Integrity Check (MIC)

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Encryption

CCMP uses AES Counter Mode for encryption.

AES is described as the most secure encryption protocol currently available and is widely used by corporations, governments, and other organizations.

AES supports multiple modes of operation. CCMP uses **Counter Mode** because it provides:

1. High performance
2. High speed

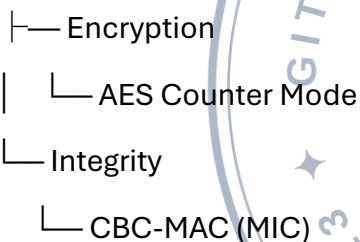
Integrity

CCMP uses **CBC-MAC (Cipher Block Chaining Message Authentication Code)** as its Message Integrity Check.

CBC-MAC is used to verify message integrity.

CCMP Structure

CCMP



Summary

AES Counter Mode --> Encryption

CBC-MAC --> Integrity (MIC)

3. GCMP (Galois Counter Mode Protocol)

GCMP is more secure and more efficient than CCMP.

Its improved efficiency allows higher data throughput.

GCMP is used in **WPA3**.

Like CCMP, GCMP combines two algorithms.

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Encryption

GCMP uses: AES Counter Mode for encryption.

Integrity

GCMP uses: GMAC (Galois Message Authentication Code)

as its Message Integrity Check.

GMAC is used to ensure message integrity.

GCMP Structure

GCMP

- └─ Encryption
 - └─ AES Counter Mode
- └─ Integrity
 - └─ GMAC (MIC)

Summary

AES Counter Mode --> Encryption

GMAC --> Integrity (MIC)

Encryption and Integrity Methods

TKIP (Temporal Key Integrity Protocol)

- Based on WEP, but more secure.
- Should not be used in modern networks.
- WPA

CCMP (Counter/CBC-MAC Protocol)

- AES counter mode for encryption
- CBC-MAC for MIC
- WPA2

GCMP (Galois/Counter Mode Protocol)

- AES counter mode for encryption
- GMAC for MIC
- WPA3

Wi-Fi Protected Access (WPA)

Wi-Fi Protected Access

Many authentication, encryption, and integrity methods exist.

To simplify deployment and create standard security combinations, the **Wi-Fi Alliance** developed WPA certifications.

WPA Certifications

The Wi-Fi Alliance created three WPA certifications:

1. WPA
2. WPA2
3. WPA3

The first certification is called **WPA**, not WPA1.

For a device to become WPA certified, it must be tested in authorized testing laboratories.

Just as devices are certified for Wi-Fi standards such as Wi-Fi 4, Wi-Fi 5, and Wi-Fi 6, they can also be certified for WPA security standards.

Authentication Modes

All WPA certifications support two authentication modes:

1. Personal Mode
2. Enterprise Mode

Personal Mode

Personal Mode uses a **Pre-Shared Key (PSK)** for authentication.

A common example is connecting to a home Wi-Fi network and entering a password.

Personal Mode is commonly used in:

1. Small networks
2. SOHO networks

Important Note

The PSK itself is not sent over the air.

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Instead, a **four-way handshake** is used.

The PSK is used to generate encryption keys.

If devices generate matching encryption keys, they can decrypt each other's traffic, confirming that both devices are using the same PSK.

Personal Mode Authentication

Client	Network
---- Four-Way Handshake ----->	
<--- Key Generation Process ----	
---- Authentication Complete -->	

Enterprise Mode

Enterprise Mode uses: 802.1X + Authentication Server

The EAP methods discussed earlier are used in Enterprise Mode.★

Examples include: PEAP and EAP-TLS

WPA itself does not require a specific EAP method.

Enterprise Mode Structure

Client	
EAP Authentication	
Authenticator	
(AP / WLC)	
Authentication Server	

WiFi Protected Access

The **WPA** certification was developed after WEP was proven to be vulnerable and includes the following protocols:

- TKIP (based on WEP) provides encryption/MIC.
- 802.1X authentication (Enterprise mode) or PSK (Personal mode)

WPA2 was released in 2004 and includes the following protocols:

- CCMP provides encryption/MIC.
- 802.1X authentication (Enterprise mode) or PSK (Personal mode)

WPA3 was released in 2018 and includes the following protocols:

- GCMP provides encryption/MIC.
- 802.1X authentication (Enterprise mode) or PSK (Personal mode)
- WPA3 also provides several additional security features, for example:

-**PMF (Protected Management Frames)**, protecting 802.11 management frames from eavesdropping/forging.

-**SAE (Simultaneous Authentication of Equals)** protects the four-way handshake when using personal mode authentication.

-**Forward secrecy** prevents data from being decrypted after it has been transmitted over the air. So, an attacker can't capture wireless frames and then try to decrypt them later.

